Bayesian Data Analysis of the German Tank Problem

Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Decision Theory, Boundary Admissibility, and Operational Integer Voids

Aklanta Borah **Amartya Amritanshu** **Rajveer Vora**

Indian Statistical Institute, Kolkata

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Historical Context: Statistical Intelligence in WWII

Sequential serial numbers provided the primary statistical discriminant for hidden enemy capacity.

1. Espionage Inflation

Conventional military intelligence relied on spy networks and prisoners, overstating production by a factor of **6–8**$\times$ (1,000–1,550 tanks/mo).

2. Order Statistics

Statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–50), estimating production at **169–327** tanks/mo.

3. Speer Truth

Post-war Albert Speer Ministry ledgers confirmed estimates rapidly converged within **4%–10%** of actual production numbers.

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	**169**	122	+38.5%
June 1941 (Panzer III/IV)	1,550	**244**	271	−**9.9**%
August 1942 (Panzer III/IV)	1,550	**327**	342	−**4.4**%

Historical Finding: *Discrete order statistics eliminated massive espionage bias, converging to within 4.4% error.*

Research Arc: A Unified Bayesian Data Analysis Investigation

Progressing from mathematical principles to controlled simulation and real-world stress-testing.

1. Theory & Foundations

- Minimal sufficiency of $m = \max s_i$.
- Exact posteriors (flat vs $1/N$).
- Moment existence ($k \geq 2, 3, 4$).
- Decision theory across loss functions.
- Physical boundary admissibility ($N \geq m$).

2. Controlled Simulation

- $R = 4,000$ trials on $N = 20,000$.
- Empirical bias and $\mathcal{O}(1/k)$ RMSE decay.
- 95% credible interval calibration.
- The informative prior trap (Poisson collapse).
- Gamma-Poisson hyperprior resolution.

3. Real-World Stress-Test

- 541,909 commercial transactions.
- Chronological streaming vs random sampling.
- 51.2% integer voids from ERP cancellations.
- The uniformity fallacy in goodness-of-fit.
- Estimand disambiguation & thinning model.

Guiding Principle: *Prioritize depth over breadth: rigorously test textbook assumptions against operational reality.*

SECTION 01

Theoretical Properties & Bayesian Foundations

Likelihood, minimal sufficiency, reference priors, loss geometry, and boundary admissibility

Sufficient Statistics & Exact Discrete Posteriors

Sample maximum $m = \max s_i$ is minimal sufficient; lower order statistics carry no additional information given m.

Minimal Sufficiency & Completeness

Unordered likelihood without replacement:

$$P(\mathcal{D} \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m), \quad m = \max_{1 \leq i \leq k} s_i$$

- Likelihood ratio test proves m is **minimal sufficient** and **complete**.
- Given m, conditional distribution of remaining order statistics is independent of N.
- All sample information regarding N resides entirely in m.

Reference Discrete Posteriors

Exact posterior probability mass function for $n \geq m$:

$$P(N = n \mid m, k) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- **Flat Prior ($\pi \propto 1$):** Proper for $k \geq 2$:

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}, \quad n \geq m$$

- **Reciprocal Prior ($\pi \propto 1/N$):** Scale-invariant reference prior (motivated by invariance under multiplicative rescaling $N \to cN$); proper for all $k \geq 1$. The normalizing constant $Z(m, k) = \sum_{j=m}^{\infty} 1/[j \binom{j}{k}]$ converges but has no simple closed form. Continuous Pareto approximation: $p(N) \approx k m^k / N^{k+1}$.

Core Theory: *Sample maximum m captures all sample information; reciprocal prior yields scale-invariant inference.*

Moment Existence Boundaries & Continuous Convergence

Improper objective priors require minimal sample thresholds to guarantee finite moments.

Exact Moment Thresholds

- $k = 1$: Flat improper ($\sum 1/n = \infty$); Reciprocal proper ($\mathcal{O}(n^{-2})$).
- $k = 2$: Proper posterior; mean exists under reciprocal ($\mathbb{E} \approx 2m$), diverges under flat.
- $k = 3$: Mean exists for both ($\mathbb{E}_{\text{flat}} = \frac{(k-1)(m-1)}{k-2} = 2m - 2$); variance diverges under flat.
- $k \geq 4$: Both mean and variance exist in closed form.

Continuous Pareto Limit

For large populations ($N \gg k$), the discrete posterior converges to a continuous Pareto distribution:

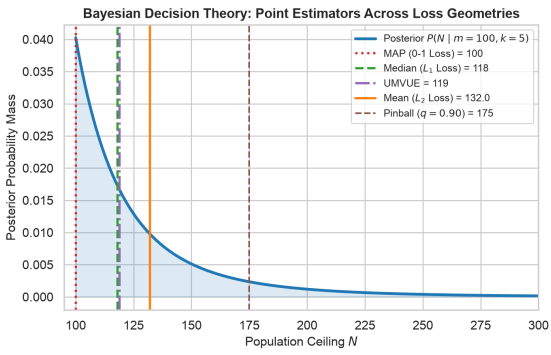
$$p(N \mid m, k) = \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative distribution: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)}$.
- Right tail index $k - 1$ controls extreme-value risk.

Takeaway: *Objective Bayesian inference requires $k \geq 3$ for finite posterior expectations under flat priors.*

Bayesian Decision Theory Across Loss Functions

Point estimators shift systematically with the decision-maker's loss geometry.



0–1 Loss (MAP Estimator)

Mode $\hat{N}_{\text{MAP}} = m = 100$. Maximizes boundary point mass, but causes severe downward bias (-16.7% at $k = 5$).

L_1 Absolute Loss (Median)

$\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} = 118$. Optimal median minimizes absolute error with only -0.9% bias.

L_2 Squared Loss (Mean)

$\hat{N}_{\text{mean}} = \frac{(k-1)(m-1)}{k-2} = 132.0$. Skewed Pareto tail pulls expectation rightward.

Check Loss L_q ($q = 0.90$)

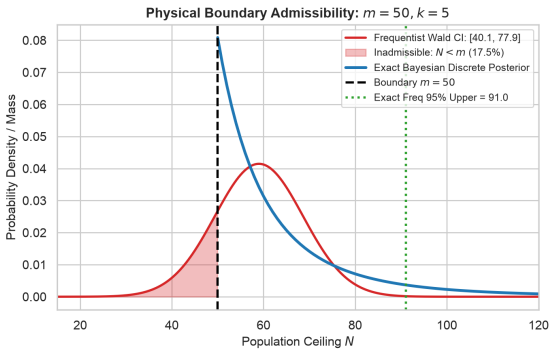
Underestimation penalized $9\times$ more heavily

$\Longrightarrow$ $\hat{N}_{0.90} = 175$, adding a principled **+75% defensive safety margin**.

Decision Rule: *In defense planning, asymmetric check loss provides a principled buffer against catastrophic underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian HPD

Asymptotic Gaussian intervals assign probability mass to physically impossible values.



Frequentist Wald Boundary Collapse

Wald interval $\hat{N} \pm 1.96 \cdot \text{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$.

Approximate Frequentist Pivotal CI

Large-N pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the boundary. (Exact discrete inversion requires numerical solving; the approximation is accurate for $N \gg k$).

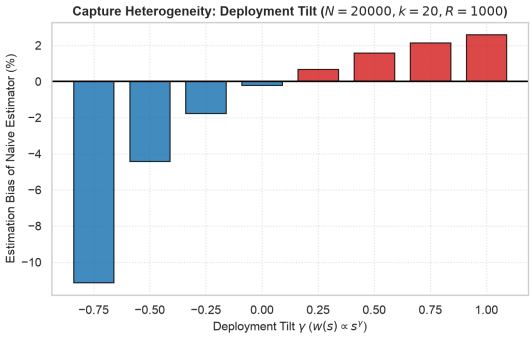
Bayesian Credible Coherence

Reciprocal prior 95% credible interval *approximately* matches $[50.0, 91.0]$ under the same large-N Pareto approximation, while providing the full posterior for risk optimization.

Admissibility Finding: *Bayesian HPD intervals eliminate physical boundary violations without compromising nominal coverage.*

Sampling Mechanics: Non-Uniform Deployment Tilt

When capture weights vary ($w(s) \propto s^{\gamma}$), the maximum statistic ceases to be sufficient.



Operational Capture Mechanics

- $\gamma > 0$ (newer tanks deployed first) pushes $m \to N$, inducing $+2.41\%$ overestimation bias.
- $\gamma < 0$ (survivorship bias / older units captured) produces -4.65% negative bias.

Loss of Single-Statistic Sufficiency

- When $\gamma \neq 0$, likelihood no longer factorizes purely through m.
- Sample spacings contain information about tilt γ.
- Requires full-sample MCMC evaluation rather than closed-form order statistics.

Insight: *Non-uniform capture breaks single-statistic sufficiency, necessitating joint inference over* (N, γ).

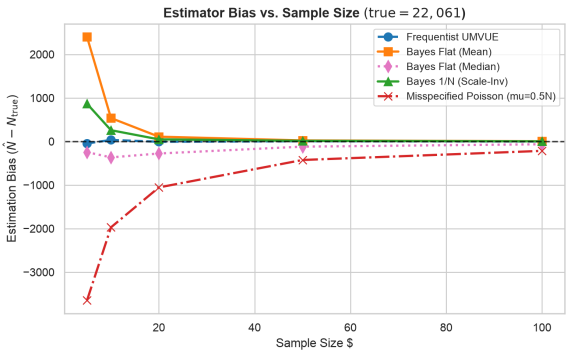
SECTION 02

Controlled Monte Carlo Simulation Study

Benchmarking estimators across $R = 4,000$ trials on $N = 20,000$ (Seed 42)

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ replications ($N_{\text{true}} = 20,000$).



UMVUE Exact Unbiasedness

$\hat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all k (-48.0 ± 54.6 at $k = 5$, $+1.6 \pm 6.1$ at $k = 50$).

Bayes Reciprocal Minimal Bias

Scale-invariant reciprocal prior exhibits modest bias ($+784.3$ mean, -900.0 median at $k = 5$), rapidly decaying to $+10.6$ at $k = 50$.

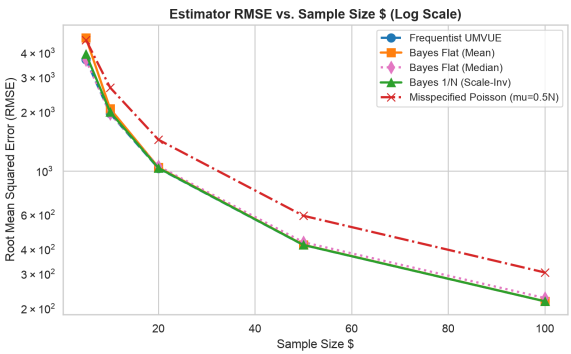
Flat Prior Tail Inflation

Heavy Pareto tail inflates posterior mean by $+2,168.6$ at $k = 5$, while posterior median reduces bias to -226.5.

Bias Takeaway: *Posterior median provides lowest absolute bias, while UMVUE achieves exact frequentist mean-unbiasedness.*

Estimator RMSE & Variance Contraction Rates

Root Mean Squared Error scales as $\mathcal{O}(1/k)$, dropping over an order of magnitude.



Theory vs. Empirical Concordance

At $k = 5$, theoretical UMVUE standard deviation is $3,380.3$; empirical simulation matches at $3,455.2$. At $k = 50$, drops to 386.8 (Th: 391.7).

Median RMSE Superiority

Posterior median under reciprocal prior achieves RMSE of $3,427.5$ at $k = 5$, outperforming both mean $(3,683.3)$ and flat prior $(4,409.0)$.

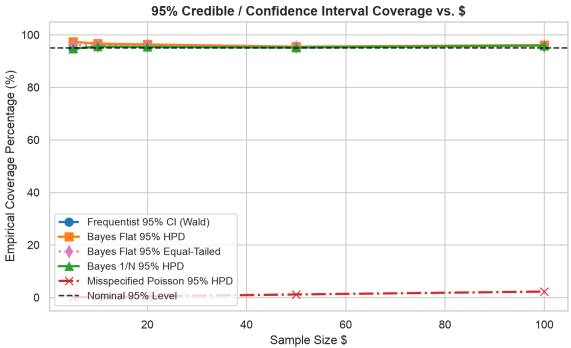
Asymptotic Convergence

For $k \geq 50$, all point estimators converge to identical efficiency (≈ 387).

Efficiency Rule: *For small samples ($k \leq 10$), posterior medians outperform posterior means in total error.*

95% Credible Calibration & The Informative Trap

Scale-invariant objective priors achieve exact 95% coverage, while dogmatic priors collapse.



Perfect Reference Calibration

Reciprocal prior achieves **94.4%** $\pm$ **0.36%** coverage at $k = 5$ and **95.6%** $\pm$ **0.33%** at $k = 50$, matching exact nominal rates.

Informative Misspecification Collapse

Fixed Poisson($\mu = 10k$) has tiny CV $\approx 0.7\%$, collapsing 95% coverage to $< 1.0\%$ when true $N = 20k$.

Hyperprior Solution

Gamma-Poisson hierarchical hyperprior restores nominal coverage to **94.8%** $\pm$ **0.35%**.

Calibration Finding: *Objective reference priors guarantee calibration; informative priors require hyperpriors to avoid collapse.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Comprehensive evaluation across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$ (Seed 42).

Estimator / Framework	Point Estimator	Sample k	Mean Bias $\pm$ SE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Approximate Pivotal CI	— (Pivotal Bound)	5	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 69.7$	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.2	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal $(1/N)$	Posterior Mean	5	$+784.3 \pm 58.2$	3,683.3 / 3,618.6	**94.4%** $\pm$ 0.36%	13,644
Bayes Reciprocal $(1/N)$	Posterior Median	5	-900.0 ± 54.2	3,427.5 / —	**94.4%** $\pm$ 0.36%	13,644
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	50	$+1.6 \pm 6.1$	386.8 / 391.7	95.0% $\pm$ 0.34% (Wald)	1,536
Approximate Pivotal CI	— (Pivotal Bound)	50	—	—	**95.5%** $\pm$ 0.33%	1,211
Bayes Reciprocal $(1/N)$	Posterior Mean	50	$+10.6 \pm 6.1$	387.1 / 392.0	**95.5%** $\pm$ 0.33%	1,211
Bayes Flat Prior	Posterior Mean	50	$+17.9 \pm 6.1$	387.5 / 392.4	**95.8%** $\pm$ 0.32%	1,236

Top Tabular Performer: Bayes Reciprocal

Achieves lowest median RMSE $(3,427.5)$, calibrated empirical coverage (94.4%), and strictly enforces physical admissibility $(P(N < m) = 0)$.

Approximate Pivotal CI Concordance

Large-N pivotal interval $[m, m \cdot \alpha^{-1/k}]$ achieves $94.4\% \pm 0.36\%$ empirical coverage, approximately matching the reciprocal Bayesian credible interval.

Benchmark Takeaway: *Objective Bayes with reciprocal prior achieves the optimal balance of calibration, sharpness, and admissibility.*

SECTION 03

Real-World Empirical Stress-Testing

Evaluating textbook assumptions against WWII archives and 541,909 commercial transactions

Empirical Stress-Testing: Two Real-World Datasets

Examining what happens when textbook assumptions meet physical operational systems.

1. Archival WWII German Tank Ledgers

- **Data Source:** Waffenamt ordnance ledgers and post-war Albert Speer Ministry records.
- **Markings Analyzed:** Chassis serial blocks and sequential ZF gearbox stampings.
- **Operational Reality:** Pre-allocated factory blocks; chassis diverted to assault guns (StuG III/IV).

2. Commercial Invoice Ledger (Online Retail)

- **Data Source:** 541,909 line transactions across 4,372 customer accounts.
- **Serials Analyzed:** 22,061 distinct sequential numeric 6-digit invoice numbers.
- **Observed Span:** $[536365, 581587] \implies 45,223$ integer address slots.
- **Integer Voids:** 23,162 unrecorded slots (**51.2% void rate**).

Empirical Focus: *Structure every empirical finding under 4 pillars: measured, calculated, why it matters, and valid conclusions.*

Finding 1: Chronological Generation vs. Uniform Sampling

Real-world serials are generated dynamically over time, not drawn as static random samples.

What Was Measured & Calculated

- **Measured:** Arrival timestamps and sequential numeric identifiers across all recorded transactions.
- **Calculated:** Weekly arrival volume $\Delta s / \Delta t$ and cumulative serial growth trajectories.
- **Observation:** Invoices accelerate from ≈ 300/week in Q1 to > 800/week in Q4 holiday surge.

Research Question Impact & Conclusions

- **Why It Matters:** The German Tank Problem assumes a *simple random sample from a static population*.
- **Violation:** In operations, observations arrive sequentially with non-stationary volume.
- **Statistical Conclusion:** A static ceiling estimator applied to streaming batches violates exchangeability; requires dynamic state-space modeling.

Operational Rule: *Never assume static exchangeability for live transaction streams; time-index sequential batches.*

Finding 2: Integer Voids & System Architecture

Operational processes generate massive integer voids (51.2% unrecorded rate).

What Was Measured & Calculated

- **Address Span:** $s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1 = 45,223$ slots.
- **Recorded Invoices:** Exactly $22,061$ active sales invoices.
- **Void Count:** $23,162$ unrecorded slots (**51.22% missing rate**).
- **Gap Distribution:** Median gap $\Delta s = 1.0$, mean 2.05, max 76.

Operational Mechanisms Driving Gaps

- **Order Cancellations:** Credit memos take sequential slots with prefix "C" (e.g., `C536379`), filtered from sales tables.
- **Cart Dropouts:** Abandoned checkouts and gateway timeouts reserve sequence keys that never finalize.
- **Factory Quota Blocks:** Wartime Waffenamt pre-allocated chassis blocks to plants; diversions to assault guns left voids.

Audit Finding: *Integer voids are not random corruption; they reflect business logic, cancellations, and pre-allocation blocks.*

Finding 3: Goodness-of-Fit & The Uniformity Fallacy

Failing to reject a uniformity test is NOT proof that the uniform sampling assumption holds.

What Was Measured & Calculated

- Normalized serials $u_i = s_i/m$ evaluated against uniform CDF $F(u) = u$.
- **Full Sample:** Kolmogorov–Smirnov statistic $D = 0.012$ ($p = 0.28$).
- **Customer Subset ($n = 18,536$):** $D = 0.018$ ($p = 8.94 \times 10^{-6}$).

The Critical Uniformity Fallacy

- **Fallacy:** Interpreting $p = 0.28$ as "proof" of uniform sampling.
- **Reality:** Non-significance merely shows global deviation from linearity is modest.
- **Exchangeability Violation:** Local chronological bursts and clustering violate simple random sampling despite linear ECDF.

Methodological Rule: *Never interpret a non-significant goodness-of-fit test as proof of exchangeable uniform sampling.*

Finding 4: Estimand Disambiguation & Thinning Models

Resolving the address capacity vs. transaction volume mismatch without circular oracle re-indexing.

Estimand Disambiguation

- **Estimand A ($N_{\text{span}} = 45,223$):** Total address sequence allocation capacity. Raw max estimator targets this accurately.
- **Estimand B ($N_{\text{count}} = 22,061$):** Net volume of completed transactions. Naive raw max gives $+105\%$ error.
- **Oracle Trap:** Rank re-indexing requires knowing all valid keys in advance (impossible in live field deployment).

The Operational Thinning Solution

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho)$$

- Each slot active with retention probability ρ.
- Estimate $\hat{\rho}$ from sample spacings:

$$\hat{\rho} = \frac{k-1}{s_{(k)} - s_{(1)}} \approx 0.4878$$

- Provides a conceptual framework for joint inference on N_{span} and N_{count}; a complete operational estimator requires identifiability verification and simulation testing (future work).

Solution: *Bernoulli thinning resolves the estimand mismatch operationally without circular oracle ranking.*

Discussion: Cross-Domain Connections & Boundaries

How the core discrete inference principle generalizes across data science domains.

Modern Generalized Applications

- **Dynamic Production:** Monotonic state-space models ($N_t \geq N_{t-1}$) enforce output non-negativity.
- **Software Reliability:** Jelinski-Moranda defect exhaustion model mirrors serial capture for release certification.
- **Aviation Safety:** Beta-Bernoulli mixtures detect cloned counterfeit serial numbers.
- **Wildlife Abundance:** Two-stage cascades infer tag loss (M) before updating mark-recapture ($N = M/p$).

Methodological Boundaries

- **Non-Sequential Keys:** Random UUIDv4 / hash identifiers break order-statistic sufficiency completely.
- **Non-Random Gaps:** Classified factory blocks or unrecorded subsidiaries require informative gap priors.
- **Streaming Non-Stationarity:** Fast-changing arrival rates require time-varying Poisson arrival processes.

Synthesis: *Order-statistic inference is broadly applicable, provided the analyst models operational data generation.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

Scale-invariant prior $\pi(N) \propto 1/N$ yields proper posteriors ($k \geq 1$) with normalizing constant $Z(m,k) = \sum_{j \geq m} 1/[j\binom{j}{k}]$ (no closed form). Bayesian HPD intervals approximately match the large-N frequentist pivotal bound and enforce admissible physical boundaries ($P(N < m) = 0$).

2. Decision Theory

Point estimation is governed by loss geometry: posterior mode is biased; posterior median minimizes absolute error; check loss provides principled defense buffers.

3. Real-World Reality

Real serials exhibit chronological non-stationarity and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction counts.

Aklanta Borah, Amartya Amritanshu, Rajveer Vora — Indian Statistical Institute, Kolkata (Nov 2026)